

Exchange Concepts Knowledge Check — Variant 02

Purpose: verify that the student has read [How a Financial Exchange Works](#) and internalized the core principles of a modern exchange, across its full scope — market history, order mechanics, risk and compliance, and technology infrastructure.

Name/ID

Instructions

- There are 30 questions.
- Each question has five statements (A-E) which may be false or true.
- Select all statements you believe to be correct by clearly filling the square before the question.
- Each question may have multiple correct statements (for some, all five statements may be correct).
- There are always at least one correct statement.

Scoring

- +1 point for each correct option selected.
- -2 points for each incorrect option selected.
- 0 points for options not selected.
- Final test score is capped at a minimum of 0: if raw total is negative, recorded score is 0.
- A passing score is 70% of the maximum rounded down to the nearest integer.

Questions

1. Debt, equity, and the IPO

- ☐ A. Bondholders are creditors with a legal claim against a company's assets
- ☐ B. A bank loan is generally better suited than a bond to raising hundreds of millions of dollars from a single lender
- ☐ C. Underwriters guarantee an IPO company its proceeds even if investor demand is weaker than expected
- ☐ D. A roadshow is used to gauge institutional demand and help set the IPO price
- ☐ E. Common stock typically carries a higher-priority claim on dividends than preferred stock

2. The three promises of an exchange

- ☐ A. Price discovery means the current price reflects the aggregate opinion of willing participants
- ☐ B. Liquidity means a participant can convert an asset to cash without an indefinite wait for a counterparty
- ☐ C. Fairness means the rules for who trades first are decided case-by-case by exchange staff
- ☐ D. Regulation NMS is the reason the US has more than a dozen competing registered equity exchanges
- ☐ E. The Securities Exchange Act of 1934 was passed before the 1929 crash

3. Participants and the maker/taker distinction

- ☐ A. An aggressive order that immediately executes against a resting order is submitted by the "maker"
- ☐ B. Payment for order flow exists partly because retail flow is more profitable to serve than institutional flow
- ☐ C. Institutional investors often use VWAP algorithms and dark pools to reduce market impact
- ☐ D. A quote is always a strictly one-sided instruction submitted by any participant
- ☐ E. Demutualisation converted many exchanges from member-owned mutuals to for-profit public companies

4. Regulators and rules

- ☐ A. FINRA is a self-regulatory organisation overseeing broker-dealers in the US
- ☐ B. The CFTC oversees US futures and derivatives exchanges
- ☐ C. MiFID II applies only to US equity markets
- ☐ D. Regulation SHO governs short sales, including the locate requirement
- ☐ E. The Market Access Rule (15c3-5) was enacted in direct response to the 2010 Flash Crash

5. Historic market crashes

- ☐ A. Black Monday, 19 October 1987, saw US markets fall 22.6% in a single day
- ☐ B. Portfolio insurance selling is cited as an amplifying factor in the 1987 crash
- ☐ C. The Brady Commission's central recommendation was to create automatic trading halts
- ☐ D. The 1929 crash was investigated and found free of manipulation or fraud
- ☐ E. The Dow fell approximately 89% from its 1929 peak to its 1932 trough

6. Instruments and the round lot

- ☐ A. A futures contract is a claim on an asset that already exists, like a share
- ☐ B. An odd lot is a quantity that is not a whole multiple of the round lot size
- ☐ C. Historically, odd-lot quotes were excluded from the NBBO and consolidated tape
- ☐ D. For most US equities, a round lot is commonly 100 shares
- ☐ E. Regulators have never revisited the exclusion of odd-lot data from public feeds

7. Stop and stop-limit orders

- ☐ A. A stop-loss order is a sell stop used to automatically exit a losing position
- ☐ B. A stop-limit order guarantees execution once triggered
- ☐ C. A plain stop order converts to a market order when triggered
- ☐ D. Stop orders are universally implemented at the exchange level on every venue
- ☐ E. A stop-limit order can fail to fill if the market gaps through the limit price

8. Trailing stop mechanics

- ☐ A. A trailing stop's stop price freezes when the market moves against the position
- ☐ B. If a \$5 trail is set and the price rises from \$150 to \$170 then falls to \$167, the stop is at \$165
- ☐ C. Before it fires, a trailing stop competes for queue position against resting limit sell orders
- ☐ D. Once triggered, a trailing stop matches directly against the best available bid
- ☐ E. The mechanism that advances a trailing stop favourably is called the ratchet

9. Hidden liquidity priority rules

- ☐ A. On some exchanges, fully displayed orders have strict priority over hidden orders at the same price
- ☐ B. The "back of queue on refresh" rule is the least common approach to iceberg replenishment
- ☐ C. A midpoint peg order never displays a price in the visible book
- ☐ D. On IEX, an aggressive order can cross to the midpoint against a resting midpoint peg
- ☐ E. Dark pools always show full pre-trade transparency of resting orders

10. OCO and combo orders in practice

- ☐ A. A bracket order combines a take-profit and a stop-loss as an OCO pair
- ☐ B. A pairs trade is an example of a combo order across two instruments
- ☐ C. Eurex offers combination option strategies such as straddles, strangles, and butterflies
- ☐ D. A calendar spread on CME involves buying and selling the same futures contract month
- ☐ E. Combo orders eliminate leg risk entirely by design

11. Implied orders

- ☐ A. Implied orders are derived from combining orders in different but related books
- ☐ B. A spread price of $-\$2.00$ means January is trading $\$2.00$ above February
- ☐ C. When an implied order matches, it consumes real orders from real underlying books
- ☐ D. Most exchanges allow unlimited depth of implied-of-implied chains
- ☐ E. Continuous recalculation of implied prices is required after every relevant book change

12. Time-in-force quick facts

- ☐ A. ATO orders are valid only during the opening auction
- ☐ B. GTD orders remain active until filled, cancelled, or their expiry date, whichever comes first
- ☐ C. An IOC order accepts a partial fill and cancels the unfilled remainder
- ☐ D. FOK orders accept partial fills as a matter of standard practice
- ☐ E. ATO orders submitted after the opening auction concludes are typically rejected

13. Depth and the order book

- ☐ A. A tight spread generally indicates a liquid, efficiently priced market
- ☐ B. The mid price is the arithmetic average of the best bid and best ask
- ☐ C. Level 2 data shows only the single best bid and ask
- ☐ D. Quantity at a price level is the sum of all resting orders' sizes at that exact price
- ☐ E. The current market price is always identical to a value stored directly on a resting order

14. Sweeping the book, numerically

- ☐ A. Given a book of 2,000@ $\$150.34$ and 1,500@ $\$150.33$ on the bid side, a 3,500-share market sell exhausts both levels
- ☐ B. After that sweep, if 3,200 shares remain at $\$150.32$, the new best bid is $\$150.32$
- ☐ C. The spread between a best bid of $\$150.34$ and best ask of $\$150.35$ is $\$0.02$
- ☐ D. The mid price between $\$150.34$ and $\$150.35$ is $\$150.345$
- ☐ E. Sweeping through multiple book levels can never produce a worse average price than the top-of-book price

15. Price-time priority and amendments

- ☐ A. Better price always takes priority over an earlier-arriving order at a worse price
- ☐ B. A cancel-and-replace always retains the original order's queue priority
- ☐ C. A quantity decrease amendment retains the resting order's original timestamp
- ☐ D. Pro-rata allocation rewards order size over arrival speed
- ☐ E. CME uses pro-rata allocation for its SOFR and Treasury futures

16. Tick size history and regulation

- ☐ A. US stocks traded in fractions of a dollar from 1792 until decimalisation around 2000–2001
- ☐ B. Decimalisation roughly doubled quoted spreads on liquid stocks
- ☐ C. Rule 612 was adopted partly to prevent re-fragmentation into sub-penny quoting after decimalisation
- ☐ D. CME's E-mini S&P 500 futures tick in increments of 0.25 index points
- ☐ E. Tick size is uniform across every venue that lists the same underlying stock

17. Rounding a midpoint to a valid tick

- ☐ A. Given a best bid of \$150.30 (tick 15030) and best ask of \$150.33 (tick 15033), the raw midpoint tick count is 15031.5
- ☐ B. That raw midpoint corresponds to a mathematically exact, tick-aligned price
- ☐ C. Rounding up in that scenario gives the extra half-cent to the seller
- ☐ D. A matching engine and a downstream P&L system must apply the identical rounding rule to avoid reconciliation breaks
- ☐ E. Round-half-to-even ("banker's rounding") is one common convention for resolving this tie

18. The matching engine's core loop

- ☐ A. The engine dequeues messages and determines whether a new order can match resting orders
- ☐ B. A fully filled order publishes a FILLED event and is removed from the book
- ☐ C. A partially filled order rests the remaining unfilled quantity in the book
- ☐ D. Checking dormant stop orders happens before any fill events are published
- ☐ E. Cancels and modifications also generate published events

19. The life of a trade

- ☐ A. Both sides of a fill receive execution reports detailing quantity, price, and remaining quantity
- ☐ B. The aggressor field only matters for regulatory reporting, not fee calculation
- ☐ C. A partial fill leaves an order resting with reduced remaining quantity
- ☐ D. Every trade is published over the market data feed to subscribers
- ☐ E. The trade record captures the IDs of both orders and both gateways involved

20. Market maker protection

- ☐ A. MMP counts fills within a configured rolling time window per market maker
- ☐ B. If MMP fires, all of that market maker's quotes are automatically cancelled
- ☐ C. MMP exists purely as a courtesy with no market-wide liquidity benefit
- ☐ D. MMP parameters are typically configured per market maker per symbol
- ☐ E. A market maker enters a protection period after MMP fires, before re-quoting

21. Auction imbalance and tie-breaking

- ☐ A. Imbalance messaging shows how many more shares are on one side than the other at the indicative price
- ☐ B. If imbalance is tied between two candidate prices, the algorithm prefers the price in the direction of remaining pressure
- ☐ C. Proximity to a reference price is used as a final tie-breaker if imbalance and pressure are both equal
- ☐ D. The indicative uncross price is recalculated only once, at the very start of the auction
- ☐ E. NYSE begins publishing closing auction imbalances at 3:45pm for a 4:00pm uncross

22. Trading session transitions

- ☐ A. PRE_OPEN transitions to OPENING_AUCTION at the scheduled auction start time
- ☐ B. HALTED can transition to RESUMPTION_AUCTION once the halt period expires
- ☐ C. CONTINUOUS can transition directly to CLOSING_AUCTION on an end-of-day signal
- ☐ D. The session state machine allows any transition the matching engine requests, without restriction
- ☐ E. CLOSED marks the point where GTC orders are persisted and the book state is saved

23. Index membership and consequences

- ☐ A. S&P 500 eligibility requires, among other things, positive earnings over the most recent four quarters and the most recent quarter
- ☐ B. Committee approval is required for S&P 500 inclusion even if all numeric rules are met
- ☐ C. Russell and FTSE index families reconstitute with full committee discretion, exactly like the S&P 500
- ☐ D. Removal from a major index tends to cause forced selling pressure
- ☐ E. Free-float adjustment excludes shares held by founders or governments that never trade

24. What indexes drive

- ☐ A. Index futures and index options settle against the published index level
- ☐ B. Index calculation is deliberately kept separate from the matching engine's critical path
- ☐ C. Stale prices can make an index lag reality when a constituent hasn't traded recently
- ☐ D. "Triple witching" refers to the simultaneous expiry of index futures, index options, and stock options
- ☐ E. The VIX is calculated directly from a company's balance sheet rather than from option prices

25. Pre-trade checks in sequence

- ☐ A. Format and syntax validation is typically the cheapest, and therefore earliest, pre-trade check
- ☐ B. Position and credit limit checks are generally cheaper to run than a rate-limit check
- ☐ C. A self-match prevention pre-check can occur before the order reaches the matching engine
- ☐ D. Running all checks in parallel is described as the preferred design for minimising latency
- ☐ E. Fat-finger price checks compare the submitted price against a cached reference price

26. Circuit breakers, design principles

- ☐ A. A minimum-halt-with-conditional-extension approach bounds uncertainty while allowing more time if needed
- ☐ B. A fixed halt duration offers participants predictability at the cost of calibration precision
- ☐ C. Larger price moves are argued to require more time for margin calls to be calculated and funded
- ☐ D. Every major exchange uses an identical halt duration regardless of instrument or move size
- ☐ E. Eurex's volatility interruptions use an auction that runs until equilibrium is found, not a fixed timer

27. Price collars and self-match prevention

- ☐ A. A static collar compares an order's price to the previous official close
- ☐ B. A dynamic collar compares an order's price to the most recent traded price
- ☐ C. Wash trading is legal as long as both resting and incoming orders come from the same participant
- ☐ D. Cancel Aggressor is described as the most common default SMP policy
- ☐ E. A kill switch leaves a participant's connection active so they can keep trading immediately

28. Mass cancel vs kill switch

- ☐ A. A mass cancel can be initiated by the exchange itself, never by the participant
- ☐ B. After a mass cancel, the participant's gateway connection remains fully active
- ☐ C. A kill switch can be triggered automatically on a lost gateway connection
- ☐ D. After a kill switch fires, reconnection and authentication are typically required before resuming
- ☐ E. Mass cancel messages generate individual cancellation events in the audit trail

29. Drop copy and the audit trail

- ☐ A. Drop copy is a public feed available to any market participant who subscribes
- ☐ B. A drop copy sequence number lets a recipient detect and request replay of missed events
- ☐ C. The audit trail must be complete, immutable, and retained for a period such as seven years under US rules
- ☐ D. Deterministic replay means the same input sequence must always reproduce the same fills and book state
- ☐ E. A clearing broker can use drop copy to begin clearing and settlement before end-of-day reports arrive

30. Knight Capital, root causes

- ☐ A. The deployment process lacked automated verification that all eight servers ran the correct code
 - ☐ B. The dangerous Power Peg code had been fully deleted from the codebase before the incident
 - ☐ C. Knight lacked a firm-wide position or notional limit that could have halted the runaway system sooner
 - ☐ D. The exchange's own systems rejected the majority of Knight's erroneous orders
 - ☐ E. Knight's kill switch, in principle available, was too slow to exercise effectively during the crisis
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